

Inban Tamizhselvan

inban.ts@gmail.com | +1 (609)-955-6401 | [GitHub Profile](#) | [LinkedIn](#) | [Website](#)

EDUCATION

University of Massachusetts Dartmouth

Jan' 2023 – Aug' 2024

Master of Science in Computer and Information Science

Rajalakshmi Engineering College, Tamil Nadu, India

Aug' 2018 - May 2022

Bachelor of Engineering in Computer Science

ACADEMIC PROJECTS

Real-time Serverless Chat Application

Jun' 2024 – Jul' 2024

University of Massachusetts Dartmouth

- Developed a real-time chat application using **React.js**, **CSS3**, and **JavaScript**, integrating **Firebase** for authentication, real-time data synchronization, and cloud storage, ensuring a responsive **user experience** with **<100ms latency**.
- Adopted **Firebase Authentication** for secure user login and registration, incorporating encrypted data storage to protect user credentials and ensure a high security compliance rate.
- Built a dynamic **user interface** with **CSS3** and **React Hooks**, applying flexible design principles to optimize usability across devices, resulting in a **35%** increase in **user engagement** and **30%** reduction in session duration.
- Leveraged **Zustand** for **state management**, optimizing asynchronous operations and state handling to ensure an uninterrupted, lag-free chat experience with real-time updates.
- Enhanced application performance through advanced data processing techniques, achieving a **50%** reduction in latency and ensuring reliable communication.

Decentralized NFT Marketplace and Token Transfer using ICP Blockchain

Jun' 2024 – Jul' 2024

University of Massachusetts Dartmouth

- Engineered a **decentralized token system** allowing users to authenticate via **ICP Internet Identity**, claim tokens through a faucet, perform token transfers, and maintain a detailed transaction history, enhancing platform security and user trust.
- Demonstrated an NFT marketplace where users can mint, buy, and sell NFTs, leveraging **inter-canister communication** to synchronize NFT transactions with the token transfer system, ensuring consistent and transparent transaction tracking.
- Implemented **Motoko-based smart contracts** to facilitate automated and secure transactions within the marketplace, reducing transaction errors and improving processing efficiency by **45%**.
- Refined the frontend architecture using **React.js**, **HTML**, **JavaScript** and **CSS3** for a smooth user interface, resulting in a **35%** improvement in user engagement and interaction rates.
- Ensured **system scalability** to handle an increasing volume of transactions and user activity, achieving **70%** higher throughput without performance degradation than other blockchains, backed by rigorous **Git version control** for seamless development and deployment.

Dynamic To-Do List Application

May 2024 – Jun' 2024

University of Massachusetts Dartmouth

- Created a **dynamic To-Do List** web application with **JavaScript**, **Embedded JavaScript (EJS)**, **Express.js**, **HTML**, and **CSS**, ensuring robust integration between frontend and backend components for a seamless experience.
- Improved **data management** by implementing **PostgreSQL**, achieving a **40%** increase in scalability and performance through efficient data storage and retrieval processes.
- Designed and deployed **RESTful APIs** to enable smooth and secure communication between the **client-side** and **server-side** systems, facilitating real-time data interactions and speeding up system responsiveness.
- Utilized **Material-UI** and responsive web design principles using **React.js** to craft a visually appealing interface, resulting in a **30%** improvement in overall user experience.

SKILLS

- **Frontend:** HTML5, CSS3, Bootstrap 5, JavaScript (ES6+), React.js, Zustand, Redux (state management), jQuery.
- **Backend:** Node.js, Express.js, EJS, RESTful APIs, GraphQL, Motoko (ICP Blockchain), Google Cloud Firestore, NPM, JWT.
- **Programming Languages:** Python (Django), Java, C, C++, PHP.
- **Databases:** PostgreSQL, MongoDB (NoSQL), MySQL (RDBMS), Firebase Database.
- **Tools:** Git and GitHub (version control), Command Line Interface (CLI), Visual Studio Code, Microsoft Office Suite, AWS.
- **Additional Skills:** Verbal Communication, Teamwork, Software Development Lifecycle (SDLC).